

Effect of cinnamon and turmeric on urinary oxalate excretion, plasma lipids, and plasma glucose in healthy subjects.



BACKGROUND: High oxalate intake resulting from consuming supplemental doses of cinnamon and turmeric may increase risk of hyperoxaluria, a significant risk factor for urolithiasis. OBJECTIVE: This study assessed urinary oxalate excretion from supplemental doses of cinnamon and turmeric as well as changes in fasting plasma glucose, cholesterol, and triacylglycerol concentrations. DESIGN: Eleven healthy subjects, aged 21-38 y, participated in an 8-wk, randomly assigned, crossover study that involved the ingestion of supplemental doses of cinnamon and turmeric for 4-wk periods that provided 55 mg oxalate/d. Oxalate load tests, which entailed the ingestion of a 63-mg dose of oxalate from the test spices, were performed after each 4-wk experimental period and at the study onset with water only (control treatment). Fasting plasma glucose and lipid concentrations were also assessed at these time points. RESULTS: Compared with the cinnamon and control treatments, turmeric ingestion led to a significantly higher urinary oxalate excretion during the oxalate load tests. There were no significant changes in fasting plasma glucose or lipids in conjunction with the 4-wk periods of either cinnamon or turmeric supplementation. CONCLUSIONS: The percentage of oxalate that was water soluble differed markedly between cinnamon (6%) and turmeric (91%), which appeared to be the primary cause of the greater urinary oxalate excretion/oxalate absorption from turmeric. The consumption of supplemental doses of turmeric, but not cinnamon, can significantly increase urinary oxalate levels, thereby increasing risk of kidney stone formation in susceptible individuals.